

A PROJECT REPORT
ON
DATA VISUALIZATION WITH PYTHON



FACULTY OF COMPUTER APPLICATIONS

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CERTIFICATE

This is to certify that **Drishti Agrawal**, a student of BCA-IBM, has successfully completed the project titled “**Fitness and Health Data Analysis using Python**” during the academic session 2025–2026.

This project has been carried out under my guidance and supervision. The work submitted is original and is in partial fulfillment of the requirements for the award of the degree.

Mr. Ferhan Ahmed

Signature

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to my project guide for their valuable guidance, support, and encouragement throughout the completion of this project on **“Fitness and Health Data Analysis using Python.”**

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I would like to extend my thanks to my friends and classmates for their support and cooperation during the development of this project.

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INDEX

<u>S.No.</u>	<u>Content Title</u>	<u>PageNo.</u>
1.	Introduction	1
2.	Problem Statement	2
3.	Objectives of the Project	3
4.	Tools and Techniques	4
5.	Dataset Description	5
6.	Practical Implementation	6–7
7.	Loading the Dataset	8
8.	Exploratory Data Analysis (EDA)	9
9.	Dataset Information	10
10.	Statistical Summary	11
11.	Insights & Visualization	12–16
12.	Recommendations for Improvement	17
13.	Conclusion	18
14.	References	19

INTRODUCTION

Fitness and health have become very important in today's modern lifestyle. People are becoming more conscious about their daily physical activities such as walking, exercising, sleeping, and maintaining a balanced routine.

With the use of fitness tracking devices and mobile applications, a large amount of data related to daily activities is generated. However, this raw data does not provide useful information unless it is properly analyzed.

Exploratory Data Analysis (EDA) is a method used to understand patterns, trends, and relationships in data. By applying EDA techniques to fitness data, we can analyze how different factors such as steps walked, calories burned, exercise duration, sleep hours, and water intake affect overall health.

This project focuses on analyzing fitness and health data using Python libraries such as Pandas, Matplotlib, and Seaborn. The aim is to convert raw data into meaningful insights through visualization and analysis, which can help in improving lifestyle and maintaining good health.

PROBLEM STATEMENT

In today's fast-paced lifestyle, maintaining good health and fitness has become a major concern. People track their daily activities such as steps walked, calories burned, sleep duration, and water intake using various devices and applications. However, this data is often available in raw form, which makes it difficult to understand patterns and draw meaningful conclusions.

Manually analyzing such large amounts of data is time-consuming and inefficient. It becomes challenging to identify trends, relationships, and important insights by simply looking at numbers in tables or spreadsheets.

To overcome this problem, data visualization techniques are used in this project. By representing the data in the form of graphs and charts, it becomes easier to understand variations in activity levels, compare different parameters, and identify patterns. Visualization helps in converting complex data into simple and meaningful insights, which can assist in improving fitness and health.

OBJECTIVES OF THE PROJECT

1. To analyze daily fitness and health data

The project aims to study daily activity records such as steps walked, calories burned, exercise duration, sleep hours, and water intake in order to understand overall fitness patterns.

2. To study the relationship between different fitness parameters

The objective is to examine how factors like steps, exercise, and sleep are connected with calories burned and overall health condition.

3. To understand patterns and trends in physical activity

By analyzing the data, the project helps in identifying whether the activity levels are consistent or vary from day to day.

4. To visualize the data using different graphical techniques

Various charts such as line charts, bar charts, pie charts, and scatter plots are used to represent the data in a clear and easy-to-understand form.

5. To identify factors affecting fitness and health

The project focuses on finding which factors contribute the most to maintaining good health and improving fitness levels.

6. To perform correlation analysis between variables

The objective is to analyze the relationship between different attributes using techniques like heatmaps to understand their dependency.

7. To draw meaningful insights from the data

The project aims to convert raw data into useful information that can help in understanding fitness behavior.

TOOLS & TECHNIQUES

In this project, various tools and techniques have been used to perform data analysis and visualization effectively.

➤ Tools Used:

- **Jupyter Notebook**

Jupyter Notebook is used as the main platform for writing and executing Python code. It allows step-by-step analysis and helps in displaying outputs such as tables and graphs in an organized manner.

- **Microsoft Excel**

Excel is used to create and organize the dataset. The data is initially stored in Excel format and then converted into CSV format for further analysis

➤ Technologies / Libraries Used:

- **Python** – Programming language used for analysis.
- **Pandas** – For data manipulation and handling.
- **NumPy** – For numerical computations.
- **Matplotlib** – For basic data visualization.
- **Seaborn** – For advanced graphical representation.

DATASET DESCRIPTION

The dataset used in this project contains daily fitness and health-related records. It is initially created in Microsoft Excel and then converted into CSV format for performing analysis using Python.

The dataset includes the following variables:

- **Day** – Represents the sequence of days for which data is recorded
- **Daily_Steps** – Total number of steps walked in a day
- **Calories_Burned** – Amount of energy expended through physical activity
- **Workout_Duration** – Time spent on exercise or physical activity (in minutes)
- **Sleep_Duration** – Total number of hours of sleep
- **Water_Intake_Liters** – Quantity of water consumed per day (in liters)
- **Activity_Level** – Categorizes activity into Low, Medium, and High
- **Day_Type** – Indicates whether the day is a Weekday or Weekend

Each row in the dataset represents the fitness data of a particular day, making it suitable for time-based analysis.

The dataset is structured and well-organized, allowing efficient data processing and visualization. It provides a clear understanding of how different lifestyle factors such as physical activity, sleep patterns, and hydration contribute to overall health.

The variation in the dataset values helps in identifying patterns, trends, and relationships between different fitness parameters.

A sample of the dataset is displayed using Jupyter Notebook to provide a better understanding of the data structure.

PRACTICAL IMPLEMENTATION

➤ Importing Necessary Libraries

```
[ ]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
sns.set(style='darkgrid')
```

1. Pandas

- **Purpose:** Core data manipulation and analysis.
- **Role:**
 - Reading data from CSV or Excel files into DataFrames.
 - Handling missing data, filtering rows, and selecting columns.
 - Performing grouping and aggregation operations.
 - Creating new columns and transforming data for analysis.
- **Why it's essential:** Pandas provides flexible and easy-to-use data structures for handling fitness datasets, allowing quick cleaning, exploration, and preparation of data.

2. NumPy

- **Purpose:** Numerical computing library.
- **Role:**
 - Performing efficient mathematical operations.
 - Supporting array-based computations.
 - Helping in calculations like averages, totals, and ratios.
 - Handling conditional operations using functions like `np.where()`.
- **Why it's essential:** NumPy improves the speed and performance of calculations and works efficiently with large fitness datasets.

3. Matplotlib

- **Purpose:** Low-level plotting library.
- **Role:**
 - Creating basic graphs such as line charts and bar charts.
 - Customizing plots with titles, labels, and styles.
 - Displaying trends in fitness data like steps and calories.
- **Why it's essential:** Matplotlib helps in visually representing data, making it easier to understand patterns in fitness activities.

4. Seaborn

- **Purpose:** Statistical data visualization built on Matplotlib.
- **Role:**
 - Creating attractive and informative statistical plots.
 - Provides built-in themes for consistent and readable visuals.
 - Easily integrates with Pandas DataFrames for group-wise visualizations and correlation heatmaps.
- **Why it's essential:** Seaborn provides advanced visualization features that help in better analysis of relationships between fitness variables.

LOADING THE DATASET

The dataset used in this project is stored in CSV format. In order to perform analysis, the dataset is loaded into the Jupyter Notebook using the Pandas library.

Pandas provides a function called `read_csv()` which is used to read the dataset and convert it into a DataFrame. A DataFrame is a structured format that allows easy data manipulation and analysis.

```
[5]: df = pd.read_csv(r"C:\Users\LENOVO\Desktop\fitness_data.csv")  
df.head()
```

```
[5]:
```

	Day	Daily_Steps	Calories_Burned	Workout_Duration	Sleep_Duration	Water_Intake_Liters	Activity_Level	Day_Type
0	Day1	4320.0	180.0	25.0	6.5	2.1	Low	Weekday
1	Day2	7560.0	310.0	42.0	7.0	2.5	Medium	Weekday
2	Day3	6890.0	275.0	38.0	6.8	2.3	Medium	Weekday
3	Day4	9120.0	410.0	55.0	6.2	2.0	High	Weekday
4	Day5	8450.0	360.0	48.0	7.3	2.7	High	Weekday

The above code loads the dataset into a DataFrame named `df`. The `head()` function displays the first few rows of the dataset, which helps in understanding its structure.

EDA (EXPLORATORY DATA ANALYSIS)

Exploratory Data Analysis (EDA) is an important step in data analysis that helps in understanding the dataset before performing any detailed analysis. It involves examining the data to identify patterns, trends, and relationships between different variables.

In this project, EDA is used to analyze the fitness dataset which includes parameters such as steps walked, calories burned, exercise duration, sleep hours, and water intake. The first step in EDA is to understand the structure of the dataset by checking the number of rows and columns and the type of data present in each column.

After understanding the structure, the dataset is checked for missing or null values. Handling missing values is important to ensure accurate results. Next, statistical analysis is performed to calculate values such as mean, minimum, maximum, and standard deviation, which help in understanding the distribution of the data.

Visualization plays a key role in EDA. Different types of charts are used to represent the data graphically, making it easier to identify trends and relationships. For example, charts help in understanding how physical activity affects calorie burn and overall fitness.

EDA also includes correlation analysis, which shows the relationship between different variables. This helps in identifying which factors have the most impact on fitness.

Overall, EDA helps in converting raw data into meaningful insights, which are useful for making better decisions and improving health and lifestyle.

DATASET INFORMATION

```
#Data Overview  
print("Data Info:")  
df.info()
```

```
Data Info:  
<class 'pandas.DataFrame'>  
RangeIndex: 29 entries, 0 to 28  
Data columns (total 8 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   Day                   15 non-null    str  
1   Daily_Steps           15 non-null    float64  
2   Calories_Burned       15 non-null    float64  
3   Workout_Duration      15 non-null    float64  
4   Sleep_Duration        15 non-null    float64  
5   Water_Intake_Liters   15 non-null    float64  
6   Activity_Level        15 non-null    str  
7   Day_Type              15 non-null    str  
dtypes: float64(5), str(3)  
memory usage: 1.9 KB
```

```
print("Missing Values:")  
df.isnull().sum()
```

```
Missing Values:  
  
Day                   0  
Daily_Steps          0  
Calories_Burned      0  
Workout_Duration     0  
Sleep_Duration       0  
Water_Intake_Liters  0  
Activity_Level       0  
Day_Type             0  
dtype: int64
```

The `info()` function provides a summary of the dataset including column names, data types, and the number of non-null values in each column.

The `isnull().sum()` function is used to check for missing values in the dataset. Initially, some missing values were found due to empty rows in the dataset. These were handled by removing the empty rows to ensure clean and accurate data for analysis.

STATISTICAL SUMMARY OF THE DATASET

```
print("\nDescriptive Statistics:")  
df.describe()
```

Descriptive Statistics:

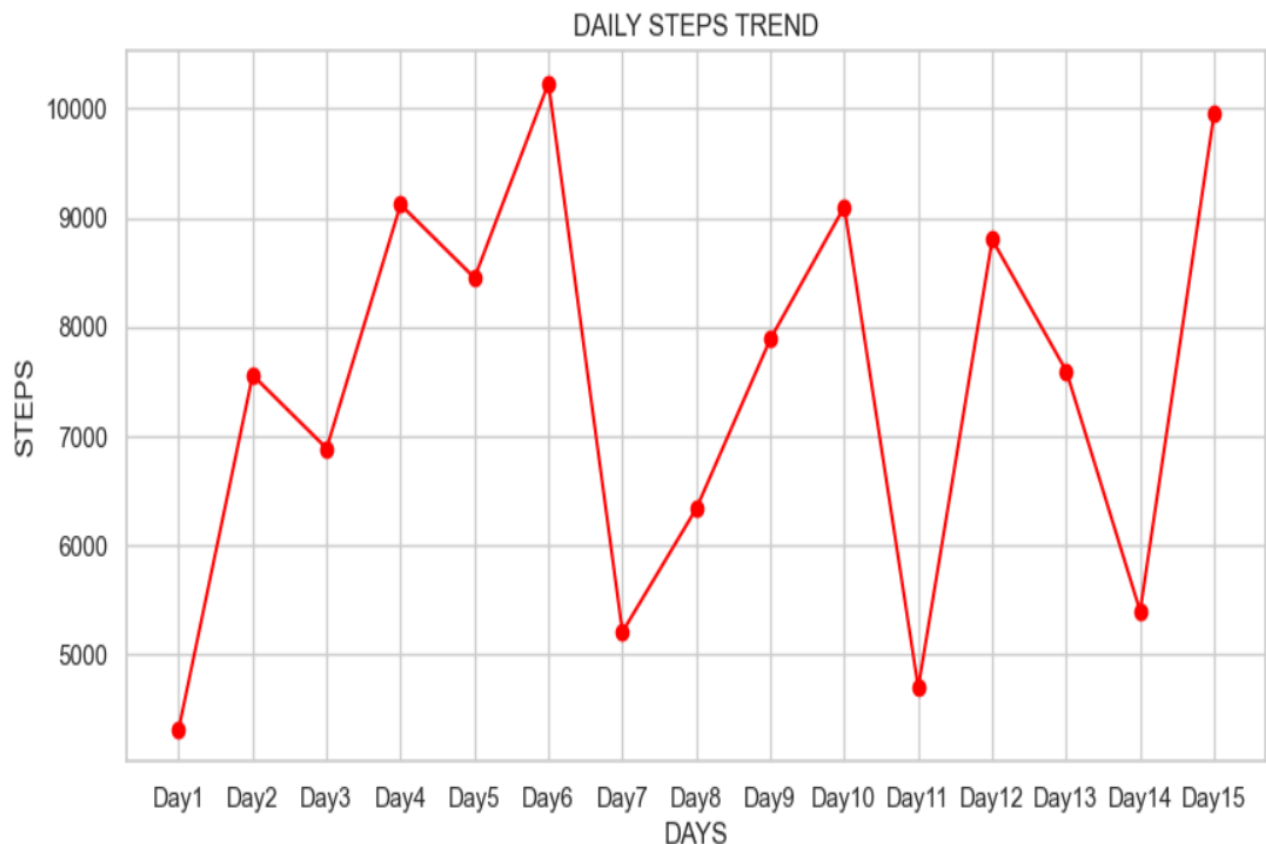
	Daily_Steps	Calories_Burned	Workout_Duration	Sleep_Duration	Water_Intake_Liters
count	15.000000	15.000000	15.00000	15.000000	15.000000
mean	7437.333333	321.000000	42.60000	7.080000	2.460000
std	1902.316132	97.489926	12.88299	0.525357	0.339748
min	4320.000000	180.000000	22.00000	6.200000	2.000000
25%	5870.000000	240.000000	32.50000	6.750000	2.250000
50%	7600.000000	320.000000	44.00000	7.000000	2.400000
75%	8950.000000	397.500000	51.00000	7.400000	2.650000
max	10230.000000	480.000000	65.00000	8.000000	3.100000

The describe() function provides statistical information about the dataset. It includes measures such as mean, minimum, maximum, and standard deviation, which help in understanding the distribution of the data.

INSIGHTS & VISUALIZATION

- INSIGHT 1 : DAILY STEPS TREND

```
plt.figure(figsize=(10,5))
plt.plot(df['Day'], df['Daily_Steps'], marker='o',color='red')
plt.title("DAILY STEPS TREND")
plt.xlabel("DAYS")
plt.ylabel("STEPS")
plt.show()
```



The line chart represents the variation in daily steps over different days. It shows that the number of steps is not consistent and fluctuates regularly. This indicates that the level of physical activity varies from day to day, with some days being more active than others.

● INSIGHT 2 : CALORIES BURNED ANALYSIS

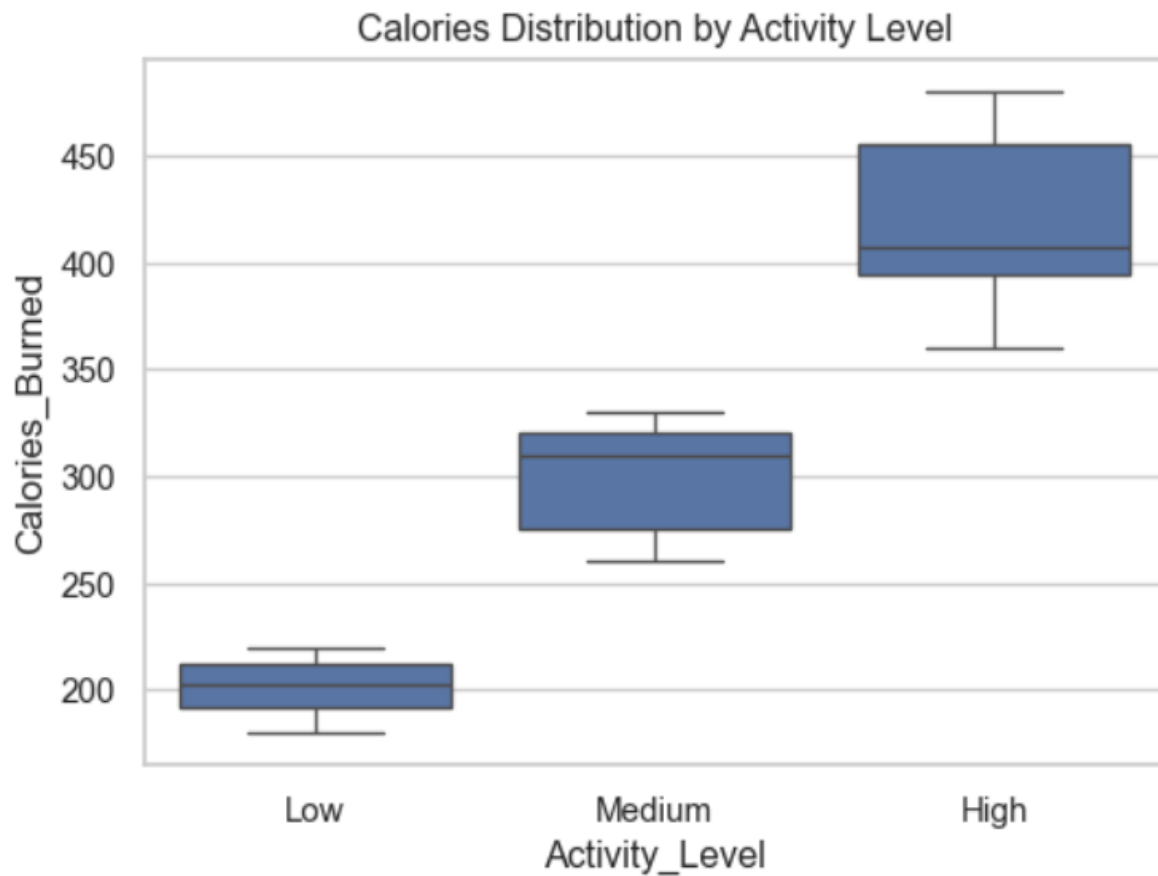
```
sns.barplot(x='Day_Type', y='Calories_Burned', data=df, hue='Day_Type')  
  
plt.title("Calories Burned: Weekday vs Weekend")  
plt.show()
```



The comparison of calories burned on weekdays and weekends shows that physical activity levels vary depending on the type of day. It indicates that individuals may be more or less active on weekends compared to weekdays, highlighting the impact of daily routine on fitness levels.

- INSIGHT 3 : ACTIVITY LEVEL & CALORIES BURNED

```
plt.figure(figsize=(6,4))
sns.boxplot(x='Activity_Level', y='Calories_Burned', data=df)
plt.title("Calories Distribution by Activity Level")
plt.show()
```



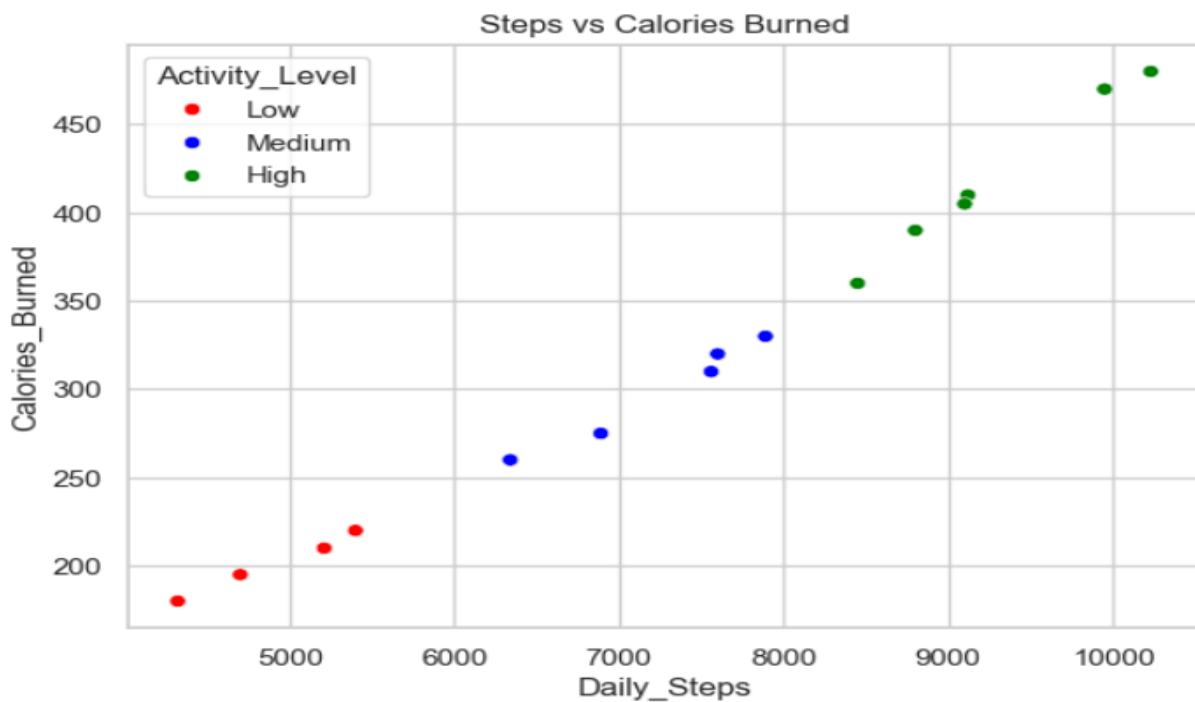
The box plot shows the distribution of calories burned across different activity levels. It indicates that higher activity levels are associated with higher calorie burn, while lower activity levels correspond to lower calorie burn.

- INSIGHT 4 : RELATIONSHIP BETWEEN STEPS AND CALORIES BURNED

```
plt.figure(figsize=(7,5))

sns.scatterplot(
    x='Daily_Steps',
    y='Calories_Burned',
    hue='Activity_Level',
    palette=['red', 'blue', 'green'], # bright colors
    data=df
)

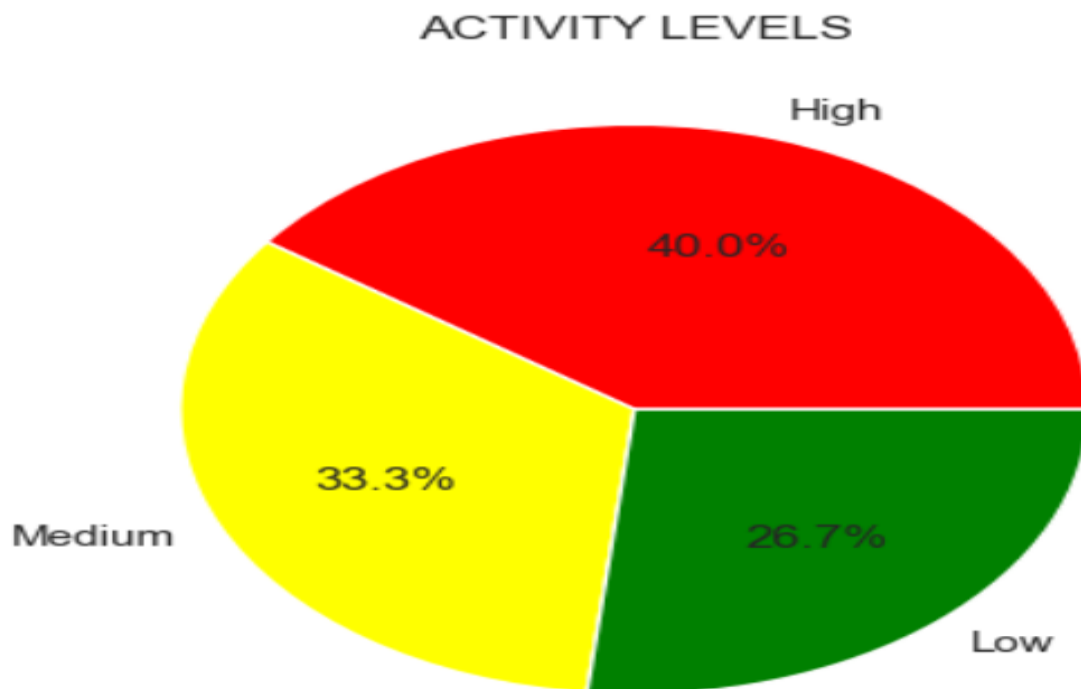
plt.title("Steps vs Calories Burned")
plt.show()
```



The scatter plot shows the relationship between daily steps and calories burned. The analysis reveals that higher physical activity leads to greater calorie burn. This highlights the importance of maintaining an active lifestyle for better health and fitness.

- INSIGHT 5: DISTRIBUTION OF ACTIVITY LEVELS

```
df['Activity_Level'].value_counts().plot.pie(  
    autopct='%1.1f%%',  
    colors=['red', 'yellow', 'green']  
)  
  
plt.title("ACTIVITY LEVELS")  
plt.show()
```



The pie chart represents the distribution of different activity levels in the dataset. It shows the proportion of low, medium, and high activity levels, helping to understand overall fitness patterns.

RECOMMENDATIONS FOR IMPROVEMENT

Based on the analysis of the dataset, the following recommendations can help in improving fitness and health:

- Maintaining consistency in daily physical activity is important. Individuals should aim to achieve a regular number of daily steps instead of having irregular activity patterns.
- Increasing workout duration can help in improving overall fitness and increasing calorie burn.
- Proper sleep is essential for maintaining good health. A balanced sleep schedule should be followed to improve energy levels and overall well-being.
- Adequate water intake should be maintained daily, as hydration plays a key role in body functioning and fitness.
- It is important to maintain a balanced lifestyle by combining physical activity, proper diet, sufficient sleep, and hydration.

CONCLUSION

The analysis of the fitness dataset shows that daily physical activity, workout duration, sleep, and water intake have a significant impact on overall health. It is observed that higher activity levels lead to increased calorie burn, while irregular routines result in fluctuations in fitness levels.

The study also highlights that physical activity is not consistent across all days, which can affect overall fitness. Maintaining a balanced routine with regular exercise, proper sleep, and adequate hydration is essential for a healthy lifestyle.

To improve fitness levels, it is important to maintain consistency in daily steps and workout duration. Individuals should aim for regular physical activity, proper sleep schedules, and sufficient water intake. Planning a structured routine and tracking daily activities can help in achieving better health outcomes.

Overall, this project demonstrates how data analysis and visualization can be used to understand fitness patterns and suggest effective ways to improve health and lifestyle.

REFERENCES

The following resources were referred to during the completion of this project:

- [Python Programming Language](#) – for implementing data analysis using Python.
- [Pandas User Guide](#) – for data manipulation and handling techniques.
- [Seaborn Visualization Library](#) – for creating advanced statistical graphs.
- [Matplotlib Tutorials](#) – for designing and customizing charts.
- [NumPy Library](#) – for numerical computations.
- Online study materials and class notes provided by the instructor.
- Self-generated dataset created in Microsoft Excel for fitness and health analysis.